PARTIDO STATE UNIVERSITY • COLLEGE OF ENGINEERING AND COMPUTATIONAL SCIENCES • BSIT CAPSTONE PROCEEDINGS (2026)

AURORA: An Academic Unified Review, Observation, Rating, and Assessment Platform for Paperless Capstone Defense Workflows

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Abstract

Traditional undergraduate capstone and thesis defense workflows in Philippine state universities rely heavily on physical manuscript printing, manual adviser endorsement routing, ad-hoc room scheduling, and hand-computed paper score sheets. These fragmented processes result in prolonged review turnaround times, scheduling collisions, high financial printing costs for students, and the absence of cryptographic auditability. This paper presents AURORA (Academic Unified Review, Observation, Rating, and Assessment), a cloud-native, paperless academic defense management and quality assurance platform engineered for the College of Engineering and Computational Sciences (CECS) at Partido State University. Built using Next.js 16 (App Router), React 19, TypeScript, and Supabase (PostgreSQL 15 with Row-Level Security), AURORA unifies four core architectural innovations: (1) a resolution-independent, dual-pane PDF.js manuscript review workspace with real-time WebSocket coordinate annotations and FIPS 180-4 SHA-256 document version fingerprinting; (2) an automated defense scheduler with venue clash detection and an academic Conflict-of-Interest Firewall that prevents research advisers from sitting as voting panelists for their own advisees; (3) a dynamic weighted rubric engine (Form DCS-CF-04) integrated with Republic Act No. 8792 step-up password re-authentication and tamper-evident digital signature serialization (AURORA-YYYY-NNNNNN); and (4) an administrative self-healing maintenance and 3-2-1 disaster recovery console. Empirical evaluation with N = 27 institutional stakeholders using a validated 10-item System Usability Scale (SUS) and 27-item Unified Theory of Acceptance and Use of Technology (UTAUT) instrument (Cronbach's $\alpha = 0.946$) yielded a composite SUS score of 91.8 / 100 (Grade A — Excellent) and a grand UTAUT mean of 4.86 / 5.00 (Highly Acceptable), while reducing manuscript endorsement turnaround time by 85.9% (from 18.4 days to 2.6 days).

Keywords: *Paperless Academic Workflow, Capstone Defense Management, Coordinate PDF Annotation, Republic Act No. 8792 Digital Signatures, Row-Level Security, System Usability Scale (SUS), UTAUT.*

1 Introduction

Undergraduate capstone research is a cornerstone curriculum requirement for computing degrees under Commission on Higher Education (CHED) Memorandum Order No. 25, Series of 2015 [1]. At Partido State University (ParSU), students enrolled in the Bachelor of Science in Information Technology (BSIT) program under the College of Engineering and Computational Sciences (CECS) progress through a rigorous four-stage oral defense lifecycle encompassing Concept Paper Defense, Title Defense, Proposal Defense, and Final Oral Defense.

Despite the digital nature of computing education, the operational administration of capstone defenses has historically remained bound to physical paper workflows. Student teams print multiple copies of 80-to-150-page manuscripts for every defense stage; research advisers annotate drafts manually with ink; defense coordinators cross-reference faculty schedules on spreadsheets to book deliberation rooms; and panel evaluators compute weighted grades manually on printed Form DCS-CF-04 sheets. Key informant interviews conducted at ParSU CECS identified four critical bottlenecks: (1) an average 18.4-day turnaround time for physical manuscript review and Form DCS-CF-03 endorsement; (2) frequent room and panelist scheduling collisions; (3) arithmetic and archival vulnerabilities in paper rubric sheets; and (4) the lack of centralized disaster recovery backups.

To resolve these challenges, this study developed and evaluated AURORA across four research objectives: (Obj. 1) digitizing the multi-stage defense workflow with SHA-256 manuscript versioning and Form DCS-CF-03 adviser endorsement gating; (Obj. 2) engineering a real-time split-screen PDF coordinate annotation workspace and conflict-free defense scheduler; (Obj. 3) implementing a dynamic weighted rubric engine (Form DCS-CF-04) secured by Republic Act No. 8792 digital signatures and a 3-2-1 Disaster Recovery Console; and (Obj. 4) evaluating system usability and technology acceptance using the System Usability Scale (SUS) [5] and Unified Theory of Acceptance and Use of Technology (UTAUT) [6].

2 Related Work

General-purpose Learning Management Systems (LMS) such as Moodle and Google Classroom support classroom assignment uploads but lack specialized academic defense governance primitives: multi-stage prerequisite gating, panel committee conflict-of-interest isolation, weighted multi-evaluator rubric consensus, and cryptographic clearance certificates. Similarly, institutional repository platforms focus on post-graduation thesis archiving rather than active, iterative pre-defense review.

Furthermore, under Philippine Republic Act No. 8792 (Electronic Commerce Act of 2000) [2] and Republic Act No. 10173 (Data Privacy Act of 2012) [3], academic records bearing electronic signatures require step-up identity verification, deterministic payload hashing (FIPS 180-4 SHA-256 [4]), and strict access control to ensure legal non-repudiation and data privacy.

3 Methodology

3.1 Agile Scrum SDLC & Four-Tier Cloud Architecture

AURORA was developed using the Agile Scrum Software Development Life Cycle (SDLC) across a 14-month academic timeline spanning Academic Year 2025–2026 through the 1st Semester of Academic Year 2026–2027 (August 15, 2025 to October 2, 2026). The platform is structured into four decoupled cloud tiers (Table 1): (1) a Client Presentation Tier built with React 19, Tailwind CSS v4, and Mozilla PDF.js; (2) an Edge & Server Action Tier running on Next.js 16 (App Router) and Vercel Edge Middleware; (3) a Backend Service Tier powered by Supabase GoTrue Auth, private S3 storage buckets, and Realtime WebSockets; and (4) a Data Persistence Tier on PostgreSQL 15 enforcing Row-Level Security (RLS) across 20 normalized relational tables.

Table 1. AURORA Four-Tier Cloud Architecture Stack

Tier	Core Stack	Governance & Security Role
1. Client UI	React 19, PDF.js	60/40 split-screen PDF & rubric drawer
2. Edge / API	Next.js 16, TS	RBAC route guard; RA 8792 step-up auth
3. BaaS Layer	Supabase Auth/S3	Private signed URLs; WAL WebSocket sync
4. Database	PostgreSQL 15	20-table RLS; atomic serials; 3-2-1 DR

3.2 Coordinate Annotations, Conflict Guard & RA 8792

When an adviser or panelist highlights text on page p, pixel coordinates (X, Y, W, H) are normalized against canvas dimensions (Cw, Ch) into floating-point percentages (x%, y%, w%, h%) ∈ [0, 100], ensuring zero positional drift across screen sizes. During defense scheduling, the server-side Conflict-of-Interest Firewall queries project_members and blocks any attempt to assign a project's Research Adviser (e.g., Prof. Froiland B. Peñalosa) as a voting panelist for their own advisee.

During oral defense grading (Form DCS-CF-04), the rubric engine computes the weighted score $Sw = \Sigma(ri \cdot wi / 100)$ across criteria summing to 100%. To lock an evaluation under RA 8792, the panelist re-authenticates their password, triggering the atomic PostgreSQL function generate_certificate_serial() (AURORA-YYYY-NNNNNN) and binding a SHA-256 signature hash to the immutable record.

3.3 Psychometric Validation & Evaluation Design

To address Objective 4, a survey instrument integrating the 10-item System Usability Scale (SUS) [5] and 27-item UTAUT model [6] (excluding redundant original items PE2 and UB2 per expert review) was validated by three institutional experts using the Good and Scates (1972) criteria, achieving a validity score of 4.89 / 5.00 (Very Highly Valid). Internal consistency across all 37 items yielded a Cronbach's Alpha of $\alpha = 0.946$ (Excellent Reliability). The instrument was administered to N = 27 ParSU CECS stakeholders (15 BSIT students, 5 advisers, 5 panelists, 1 coordinator, and 1 college dean).

4 Results and Discussion

4.1 Operational Efficiency & QA Verification

Transitioning from physical paper routing to AURORA's cloud workflow reduced average manuscript review and Form DCS-CF-03 endorsement turnaround time from 18.4 days to 2.6 days—an 85.9% improvement (Table 2). Automated room and panelist conflict detection eliminated double-booking incidents (0.0% clash rate), while all 42 functional, RLS security, and signature immutability test cases passed (100.0% pass rate). Additionally, the Maintenance & Disaster Recovery Console (/admin/system-health) provides 6 live diagnostic probes, one-click self-healing pointer repair, and SHA-256 verified 20-table JSON cold archives (RTO ≤ 4 hours).

Table 2. Operational Comparison: Paper Workflow vs. AURORA

Operational Metric	Paper Process	AURORA Platform
Endorsement Turnaround	18.4 days	2.6 days (−85.9%)
Manuscript Pages Printed	1,800+ pages	0 pages (Paperless)
Rubric Consolidation	45–60 mins	< 1.0 sec (Instant)
Room / Panel Clashes	14.2% rate	0.0% (Hard Guard)
DR Backup RTO / RPO	None (Manual)	≤ 4h RTO / WAL PITR

4.2 Objective 4: Empirical SUS & UTAUT Results

Table 3 presents the empirical evaluation results (N = 27). AURORA achieved a composite SUS score of 91.8 / 100 (normalized mean $\mu = 4.84$, SD = 0.28), earning a Grade A (Excellent) usability rating. Across the six UTAUT constructs, Behavioral Intention rated highest ($\mu = 4.90$, SD = 0.20), followed by Performance Expectancy ($\mu = 4.88$, SD = 0.22), Use Behavior ($\mu = 4.87$, SD = 0.24), Effort Expectancy ($\mu = 4.86$, SD = 0.25), Facilitating Conditions ($\mu = 4.85$, SD = 0.26), and Social Influence ($\mu = 4.81$, SD = 0.30), yielding a Composite Grand Mean of 4.86 / 5.00 (Highly Acceptable).

Table 3. Empirical SUS & UTAUT Evaluation Summary (N = 27)

Evaluation Construct	Mean	SD	Verbal Rating
Part I: SUS (10 Items)	4.84	0.28	Grade A (91.8/100)
PE: Performance Expectancy	4.88	0.22	Highly Acceptable
EE: Effort Expectancy	4.86	0.25	Highly Acceptable
SI: Social Influence	4.81	0.30	Highly Acceptable
FC: Facilitating Conditions	4.85	0.26	Highly Acceptable
BI: Behavioral Intention	4.90	0.20	Highly Acceptable
UB: Use Behavior	4.87	0.24	Highly Acceptable
Composite Grand Mean	**4.86**	**0.25**	**Highly Acceptable**

5 Conclusion and Recommendations

AURORA successfully modernizes the undergraduate capstone defense lifecycle at Partido State University by integrating coordinate-pinned PDF review, conflict-free scheduling, dynamic weighted rubric scoring, RA 8792 cryptographic signatures, and 3-2-1 disaster recovery into a unified cloud platform. With an 85.9% reduction in review turnaround time, a 91.8/100 SUS score, and a

4.86/5.00 UTAUT acceptability mean, the system is empirically validated for full institutional deployment across ParSU colleges.

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References

[1] Commission on Higher Education (CHED). 2015. CMO No. 25, s. 2015: Revised Policies, Standards, and Guidelines for BSCS, BSIS, and BSIT Programs.

[2] Congress of the Philippines. 2000. Republic Act No. 8792: Electronic Commerce Act of 2000.

[3] Congress of the Philippines. 2012. Republic Act No. 10173: Data Privacy Act of 2012 and NPC Circular No. 16-01.

[4] National Institute of Standards and Technology (NIST). 2015. Secure Hash Standard (SHS), FIPS PUB 180-4.

[5] John Brooke. 1996. SUS: A 'Quick and Dirty' Usability Scale. In Usability Evaluation in Industry, Taylor & Francis, 189–194.

[6] Viswanath Venkatesh, Michael G. Morris, Gordon B. Davis, and Fred D. Davis. 2003. User Acceptance of Information Technology: Toward a Unified View. MIS Quarterly 27, 3 (2003), 425–478.

[7] ISO/IEC 25010:2011. Systems and Software Engineering — Systems and Software Quality Requirements and Evaluation (SQuaRE).